

HotButton

Version 2.0.0 User Manual

Purpose

Setup, operation, Companion integration, LED control, multi-device overview, recovery and troubleshooting.

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Product	HotButton for Bitfocus Companion
Public documentation	github.com/Thomosaurus6/HotButton
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This manual describes user-visible behavior. Source code, hardware design, component details, wiring and manufacturing information are outside its scope.

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1. Quick Start

This chapter takes a new HotButton from power-up to a working physical button and a visible LED result in Companion. Complete the steps in order; later chapters explain every setting in detail.

1.1 What you need

- A HotButton running firmware 2.0.0.
- A network connection that supplies the device with power and network access.
- A computer running Bitfocus Companion with the matching HotButton connection module.
- The IPv4 address or individual local network name of the HotButton.

1.2 First connection in seven steps

Step	Action	Expected result
1	**Connect the HotButton.** Wait while it requests a network address.	Red breathing changes to a green confirmation flash when the network is ready.
2	**Open the device.** Use its IP address or individual **.local** name in a browser.	The Home page shows Device overview and Network configuration.
3	**Set the Companion address.** Save the IPv4 address of the Companion computer on the Home page.	The device can contact Companion for Surface and OSC pairing.
4	**Check Surface connection.** Leave it enabled in Settings and map the one-key Surface in Companion.	A physical press can operate the mapped Companion button.
5	**Add one HotButton connection.** Select Learn mode and press the physical button.	The connection learns the device identity and address.
6	**Create an LED test.** Add LED Color, LED Brightness and LED State actions.	The button displays the selected color at the selected brightness.
7	**Verify monitoring.** Add Device Online and Button Press feedbacks.	Companion shows availability and responds to presses.

Recommended first LED test

Set LED Color to a preset, LED Brightness to 40%, LED Flash Mode to Off and LED State to On. These parameters are separate, so a color or brightness action alone does not automatically switch the LED on.

1.3 If the device is not reachable

DHCP is the default. If no lease arrives within 60 seconds, HotButton uses the recovery address **192.168.1.10/24**. Temporarily place the computer in the same subnet, open **http://192.168.1.10/**, correct the network settings, and then restore the computer's normal configuration.

Multiple unconfigured devices

Connect and configure one new device at a time. Several devices using the same recovery address cannot be accessed reliably until each receives a unique address.

2. What HotButton Provides

HotButton is a network-connected illuminated hardware button for Bitfocus Companion. It combines a physical input, direct Surface registration, OSC-based control and monitoring, and a browser interface for local administration.

Area	Capabilities
Physical control	Immediate Press and Release events; configurable Long Press behavior; one-key Companion Surface.
LED output	Preset and custom RGB colors, transitions, brightness, power control, ten preset flash modes and a configurable Custom pulse.
Companion integration	Actions, feedbacks, variables, Learn pairing, Manual addressing and online monitoring.
Browser administration	Network settings, device name, Surface enable switch, brightness limits, LED Override, diagnostics, restart and firmware update.
Multi-device operation	Individual local names and the shared read-only overview at hotbutton.local.
Recovery	DHCP fallback address, boot reset gesture, diagnostics and browser-based firmware update.

2.1 Two independent Companion paths

HotButton uses two related but independent paths. Understanding this distinction prevents most setup confusion.

Path	Purpose	Configured where
Surface connection	Registers the physical key directly as a one-key Companion Surface.	Companion IP on the device Home page; enable switch in browser Settings.
HotButton connection module	Pairs the device for LED actions, feedbacks, variables and online monitoring.	A separate HotButton connection in Companion, using Learn or Manual mode.

Important

Surface can be disabled while module-based LED control and monitoring continue. Likewise, successful module pairing does not by itself create or map the Surface.

3. Device Identity, Names and Addresses

3.1 Device identity

Every physical HotButton has a permanent device identity. Renaming the device changes its display name and local hostname, but it does not change the hardware identity used for pairing.

3.2 Display name and individual local address

The browser Settings page allows a free display name and can restore the original generated name. The individual local address is derived from the display name and keeps a unique suffix, for example **glowing-jaguar-4b6c.local**. Use unique display names to make a multi-device installation easy to understand.

3.3 IP address and hotbutton.local

- **Device IP:** the most direct browser address.
- **Individual .local name:** follows one physical device and normally survives DHCP address changes.
- **hotbutton.local:** a shared read-only overview hosted by one active HotButton.
- **Device /overview:** a local overview endpoint available on every unit, useful while the shared name changes host.

Local names require multicast name resolution. Router, operating-system and browser caches can briefly delay a name change. When a local name is unavailable, use the current IP address.

4. Network Setup and Device Access

4.1 DHCP

DHCP is the default and is recommended for initial setup. The network assigns the address; the device Home page and Companion connection show the active IP.

4.2 Static IPv4 configuration

A static address, subnet mask and gateway can be saved on the Home page. A learned, paired Companion connection can also send static network settings and then switch itself to Manual mode while retaining the device identity.

A gateway value of **0.0.0.0** disables gateway ping monitoring. This does not disable normal heartbeat-based online monitoring.

4.3 DHCP fallback

Item	Value
Delay	60 seconds without a DHCP lease
Recovery address	192.168.1.10
Subnet	255.255.255.0 (/24)
Gateway	0.0.0.0

4.4 Other IP ranges

The shared overview can discover announcements from other IPv4 subnets on the same Ethernet network when multicast reaches them. Discovery does not create routing. To open or reconfigure a listed device, the computer still needs an address or route that can reach that device.

4.5 Access checklist

- Confirm the link is active and the device reaches Network Ready.
- Try the active IP address before troubleshooting a local name.
- Check for duplicate static or fallback addresses.
- Disable client isolation or multicast filtering when local discovery is required.
- Configure new fallback devices one at a time.

5. Companion Setup and Pairing

5.1 Add the connection

Create one HotButton connection for each physical unit. The connection provides LED actions, feedbacks, variables and online state.

5.2 Learn mode

Select Learn and press the intended physical HotButton. An intentional press starts the pairing handshake. Heartbeat traffic alone cannot claim an unpaired connection. Only the expected sender address and device identity can confirm the pair.

Use **Forget current pairing and wait for next button press** when a Learn connection must be assigned to another physical unit.

5.3 Manual mode

Enter the HotButton IP address. The connection validates the device identity. Traffic from another address or identity is rejected.

5.4 Pairing and saved settings

- A normal restart preserves pairing.
- Saving a different Companion IP on the device clears its OSC pairing so it can establish the new relationship.
- Saving Companion IP 0.0.0.0 clears that target and returns to discovery.
- A confirmed connection is not silently reassigned to another device during the same boot.

5.5 Online monitoring

The firmware heartbeat interval is 2000 ms. The module Online Timeout defaults to 10000 ms. If no valid traffic arrives within the timeout, the connection becomes offline and clears stale pressed and Long Press state.

Recommended Online Timeout

Use 10000 ms for a responsive offline indication with enough tolerance for ordinary network jitter. Increase it only when the network regularly delays valid heartbeat traffic.

6. Surface Connection

When enabled, HotButton connects directly to Companion's Satellite service and registers one physical key as a Surface. Set the Companion IP on the Home page, enable Surface connection in Settings, and map the new Surface button in Companion.

6.1 Enable and disable

- Disabling Surface stops key events immediately and closes the independent Surface connection.
- OSC pairing, LED actions, feedbacks and variables remain active.
- Re-enabling starts reconnection; discovery and registration can take several seconds.
- If the key was held while disabling, the device queues a release before closing the connection.

6.2 Migrating from an older Surface integration

The current firmware registers itself directly. Remove or disable an older separate Surface bridge to avoid stale or duplicate entries. Review button mappings because assignments stored against an older Surface may need to be transferred.

7. Browser Interface

7.1 Home page

Open the device IP or individual local name. Home shows Device overview and Network configuration, including separate Surface and OSC pairing states.

7.2 Settings page

Setting	Default / range	Behavior
Device name	Current name	Changes display name and individual local hostname; identity remains unchanged.
Restore original name	Original generated name	Selects the saved original name for the next Save settings operation.
Surface connection	Enabled	Enables or disables direct Surface registration.
Status brightness	17% if not previously saved	Controls network, pairing and reset indication brightness.
Maximum LED brightness	60%	Scales all visible LED output, including status and override.
Search indicator duration	5 minutes; 0-60 or Always	Controls only the blue/orange search light; discovery continues after it turns off.
LED Override	Disabled	Displays a fixed test color and brightness while normal logic continues underneath.

Save settings persists changes and returns to Home after success. Validation errors keep the form open. Changing the search duration uses elapsed time from the original search start and does not restart the timer.

7.3 LED Override and color picker

Enable LED Override to use the color picker, RGB fields and brightness control. RGB values are constrained to integers from 0 to 255. The picker and numeric values remain synchronized.

- Live previews are stored in RAM and do not write flash.
- An unsaved preview reverts when leaving the page or after about 30 seconds without refresh.
- A saved enabled override survives restart and masks status and Companion LED output.
- Pairing, networking, button events and underlying OSC LED state continue while Override is visible.
- Disabling Override immediately reveals the current normal LED state.

> **Copy a picked color to Companion**
>
> Use LED Color - Custom RGB with the displayed values, set LED Brightness to the displayed percentage, set LED Flash Mode to Off, set LED State to On, then disable and save LED Override.

7.4 Restart, diagnostics and update

Settings provides Restart device, Download diagnostics and Firmware update. Diagnostics are read-only operating information used for troubleshooting.

8. Understanding LED Control

HotButton treats color, brightness, flash mode and power as separate state. This makes complex Companion logic possible, but it also means a color change does not automatically switch the LED on.

Layer	What it controls
LED Color	Target RGB value and optional transition duration.
LED Brightness	The normal upper output level used by solid and flashing output.
LED Flash Mode	Solid output, preset flash 1-10, or Custom pulse.
LED State	Whether the stored normal LED output is visible.
Maximum LED brightness	Browser-level safety/output cap applied after normal brightness.
Status brightness	Separate level for startup/network/pairing indications, also scaled by Maximum.

Layer	What it controls
LED Override	Fixed browser test output that masks normal and status output.

8.1 Brightness example

With LED Brightness 80% and browser Maximum LED brightness 60%, the visible result is capped below full hardware output. Changing Status brightness does not change Companion-controlled brightness.

8.2 LED Reset

LED Reset sets the normal OSC state to white, 0% brightness, flash Off and LED power Off. It does not disable a browser LED Override.

9. Companion Actions

Action	Inputs and behavior
LED Color	Preset or Custom RGB (0-255 each). Optional transition 0-60000 ms. A new command during a fade starts from the current interpolated color.
LED Brightness	Set, Increase or Decrease. Values are constrained to 0-100%. This value is also the Custom pulse upper level.
LED Flash Mode	Off selects solid output. Presets 1-10 run from slowest to fastest. Custom resumes the current session's custom settings.
LED Flash Custom	High and Low Phase 10-60000 ms, Fade In and Fade Out, Minimum and Maximum Brightness 0-100%.
LED State	On, Off or Toggle. Off hides output without discarding color, brightness or flash mode.
LED Reset	White, brightness 0%, flash Off and LED power Off.

9.1 Color transitions

Fade duration 0 changes color immediately. A positive duration interpolates from the current visible color state to the selected target. A later command interrupts the current fade and starts from its current interpolated value.

Example: White to Red over 500 ms, then Red to Green over 500 ms, then Blue at 0 ms. Use a Companion Internal Wait at least as long as the fade when the next action must wait for completion.

9.2 Custom pulse

LED Flash Custom sets the pulse timings and sets the shared LED Brightness to its Maximum Brightness. Later LED Brightness actions change that same upper level; they do not apply an additional multiplier.

Parameter	Meaning
High Phase	Total time at the upper part of the cycle, including Fade In.
Low Phase	Total time at the lower part of the cycle, including Fade Out.
Fade In	Rising transition within High Phase; limited to that phase duration.
Fade Out	Falling transition within Low Phase; limited to that phase duration.
Minimum Brightness	Configured lower level, limited so it cannot exceed current LED Brightness.
Maximum Brightness	Sets the shared LED Brightness when the Custom action runs.

Example: Minimum 20 and Maximum 100 pulses from 20 to 100. A later LED Brightness action of 40 changes the effective range to 20-40. Brightness 5 makes both effective levels 5, so the pulse appears paused. Raising brightness restores the pulse.

After restart, Custom defaults are High 1000 ms, Low 1000 ms, Fade In 1000 ms, Fade Out 1000 ms and Minimum 20%. The upper level is the current LED Brightness. Custom parameters are session state and are not written to device settings.

9.3 Value and expression modes

Control	Accepted value
LED Flash Mode	0 = Off, 1-10 = presets, 11 = Custom
LED State	on, off, toggle
RGB	0-255 per channel
Brightness	0-100
Color fade	0-60000 ms
Custom phases	10-60000 ms

Use numeric variables for Flash Mode and the exact lower-case text for LED State. Invalid expression results send no command. The Companion Connection selector remains a fixed dropdown and cannot be supplied by a variable.

10. Feedbacks, Long Press and Variables

Feedback	Behavior
Button Press Pulse	True for the configured hold time after each physical press. Release does not shorten the pulse.
Long Press Event	Becomes active at the threshold while held, or produces a one-second pulse on release, depending on connection settings.
Device Online	True while valid traffic was received within Online Timeout.
LED State	Reports stored normal LED power state.
LED Color	Uses stored normal OSC RGB values as the feedback color.
LED Flash Mode	Matches Off, 1-10 or Custom; false while offline.

10.1 Long Press

A normal Press is sent immediately. Long Press is an additional Companion-side event and does not replace Press. The default threshold is 2000 ms. In held mode, the Long Press variable and feedback remain true until Release, then return to false and are ready for the next hold.

A heartbeat that reports Released can recover from a lost UDP release and clear stale held or Long Press state.

10.2 Variables

Group	Variables
Device	device_id, device_name, device_ip, pairing_state, online, last_seen
Button	button_state, long_press, last_event, last_press_duration_ms, last_press_duration_seconds, last_press_duration_minutes, last_press_interval_ms, last_press_interval_seconds, last_press_interval_minutes
LED	led_state, led_color, led_brightness, led_flash

LED variables describe the stored OSC state. They do not report the browser LED Override. The led_flash variable displays Off, 1-10 or Custom.

11. Multi-device Overview

Open **http://hotbutton.local/** for a read-only list of active HotButtons. The overview shows device name, individual local address, IP address, Companion IP, firmware version and Last Press time. Device names open the individual browser pages.

11.1 Overview host election

- The first available eligible HotButton hosts the shared name.
- If that host disappears, another active unit takes over after approximately ten seconds plus a short name check.
- A returning former host does not displace the current host.
- Overview membership is rebuilt after restart and is not stored as permanent event history.

11.2 Temporary name delays

Router, browser and operating-system mDNS caches can briefly delay access after host changes. Use the current host IP with **/overview** while waiting. An unrelated device that already owns hotbutton.local causes a conflict notice; individual names and IP addresses continue to work.

11.3 Last Press

Last Press shows only the time of the most recent physical press seen since startup. A dash means no press has been observed in the current runtime.

12. Startup and Status Indicators

State	Indication	Notes
No usable IP	Red breathing	Continues while the device has no usable address.
Network ready	Green fast flash for 1.5 s	Confirms DHCP or static networking.
DHCP fallback	Violet fast flash for 1.5 s	Recovery address 192.168.1.10/24 is active.
Checking saved Companion	Blue breathing	Connection checks continue every five seconds.
Unpaired or recovering	Orange breathing	Discovery and checks continue.
Connection confirmed	Green fast flash for 1.5 s	Then normal Companion-controlled LED output resumes.
Boot reset confirmation	Red fast flash for 1.5 s	Release during the reset window to clear network/pairing data.

The blue/orange indication uses one configurable window beginning at the first connection search state, not necessarily at power-on. Default duration is five minutes. When it expires, only the visible search light stops; discovery, connection checks and button presses continue.

Override warning

An enabled LED Override masks startup, search and reset indications. Disable Override when diagnosing status colors.

13. Recovery, Updates and Diagnostics

13.1 Boot reset

Hold the physical button before applying power. Keep it held through the five-second boot recovery threshold and release during the red confirmation indication. The device clears saved network and pairing configuration and restarts with DHCP and discovery.

The display name and browser preferences remain stored. If DHCP still fails, the normal recovery fallback activates after 60 seconds.

13.2 Firmware update

Step	Action
1	Open Settings and select Firmware update.
2	Upload the matching HotButton firmware update file.
3	Wait for verification to finish.
4	Select Install and restart.
5	Keep power connected until the device returns, then confirm Firmware 2.0.0.

Do not interrupt installation

Removing power during installation can leave the device requiring a local recovery procedure.

13.3 Diagnostics

Download diagnostics from Settings before changing several settings at once. For support, record the firmware version, device name and ID, current IP, Companion IP, and whether the problem affects Surface, module pairing, LED output or all communication.

14. Troubleshooting

Symptom	Likely cause	What to do
Device page does not open	Unknown address, wrong subnet or local-name cache	Try the current IP. After 60 seconds without DHCP, try 192.168.1.10 from the same /24 subnet.
hotbutton.local is unavailable	Overview host changed or mDNS cache is stale	Wait briefly, then use a known device IP with /overview.
Device appears in overview but cannot be opened	It is in another IPv4 range	Temporarily give the computer a reachable address/route, correct the device network settings, then restore the computer.
Orange indication remains	No confirmed OSC pairing	Create a Learn connection and press the device, or verify Manual address and Companion IP.
Surface is missing	Wrong Companion IP, Surface disabled or registration still reconnecting	Verify Home/Settings, wait several seconds, and check the Surface mapping in Companion.
LED actions appear to do nothing	LED State Off, brightness 0, or color/brightness sent without enabling output	Set Color, Brightness, Flash Off and LED State On together for a test.
Visible LED differs from feedback	Browser LED Override is active	Disable and save Override to reveal the underlying OSC state.
Custom pulse looks constant	Current LED Brightness is at or below Minimum	Raise LED Brightness or lower the Custom minimum.
Device remains online too long after disconnect	Online Timeout is high	Use the recommended 10000 ms unless the network needs more tolerance.
Release or Long Press appears stuck	A release packet was lost and heartbeat recovery has not arrived	Check network traffic; the next Released heartbeat normally clears the state.
Renamed device no longer opens by old name	Hostname changed with display name	Use the current IP or the new individual local name.
Status colors are not visible	LED Override masks status output	Disable Override before diagnosing startup or reset indications.

15. Reference

15.1 Default values

Item	Default
Firmware / module release	2.0.0
OSC UDP port	13122
Surface service	Companion Satellite TCP 16622
Heartbeat interval	2000 ms
Online Timeout	10000 ms
DHCP fallback delay	60 seconds
Fallback address	192.168.1.10/24
Fallback gateway	0.0.0.0
Status brightness	17% when no value has previously been saved
Maximum LED brightness	60%
Search indicator duration	5 minutes
Long Press threshold	2000 ms
Custom High / Low	1000 ms / 1000 ms
Custom Fade In / Fade Out	1000 ms / 1000 ms
Custom Minimum	20%

15.2 Input ranges

Input	Range
RGB channel	0-255 integer
LED and Override brightness	0-100%
Color transition	0-60000 ms
Custom High / Low Phase	10-60000 ms
Custom Fade In / Fade Out	0 up to the respective phase duration
Search indicator duration	0-60 minutes or Always

15.3 Practical configuration recipes

Result	Actions
Solid color	LED Color; LED Brightness; LED Flash Mode Off; LED State On.
Smooth color change	LED Color with a non-zero Fade duration; keep LED State On.
Preset flash	Set Color and Brightness; choose Flash Mode 1-10; LED State On.
Custom pulse	Run LED Flash Custom with timing/minimum/maximum; LED State On.
Temporarily hide LED	LED State Off. Use On later to restore stored color and animation.
Return to known off state	LED Reset.

16. Support, Availability and Scope

Interested in HotButton, integration or availability? Contact **Thomas Thielen**.

The public HotButton repository contains product and user documentation. Source code, firmware packages, Companion installation packages, schematics, component details, wiring and manufacturing information are not part of the public documentation.

Before requesting support

Download diagnostics and note the firmware version, device ID, device IP, Companion IP, connection mode, Online Timeout and the exact status indication. Also state whether LED Override is enabled.

Public documentation: github.com/Thomosaurus6/HotButton

End of manual